

CS 5330 HW8

1. **What method is used to calculate optical flow when motions are large? How much extra computation does it take?**

Image pyramids. This requires scaling down the image to a size where the sobel filters can be applied, and scaling back up the optical flow image to the original size.

2. **Calculating Harris corners requires calculating the eigenvalues of what?**

The gradients of a location/feature in the image that we want to look at to know if there is motion

3. **Give one example of the aperture problem when identifying good features to track.**

The aperture problem says if you are looking at flat edge of an object, for example, you can only tell that the object is moving in the direction normal to that edge. Likewise, you could also say that rotation cannot be determined on a uniform color sphere rotating.

4. **What is a background model in the context of a stationary camera?**

A background model represents the state of the world when no moving objects are present. Each pixel will contain the value that is most consistent across time. This helps identify moving or static objects, which get filtered out.

5. **What would be the result of non-maxima suppression on the following sequence of numbers?**

1 3 5 4 5 7 8 9 5 6 7 4 2 1:

0 0 5 0 0 0 0 9 0 0 7 0 0 0

6. **Give some examples of where non-maxima suppression is useful.**

Non-maxima suppression is useful in calculating the best corner locations to use as a feature in Harris Corner detection. It helps filter out pixels in an image that are not corners or have weaker "ideal" corner properties like perpendicular edges / two strong eigenvalues.

descriptors of image locations that are good features to track (rotation, scale, intensity invariant)

7. What method does SURF use to speed up the feature calculations relative to SIFT? What approximation does SURF make to enable the speedup?

It used box filters to approximate gaussians / Sobel filters. When using box filters to create integral images, they allow the pixel sum of any arbitrary rectangle to be computed much more efficiently.

8. What is the purpose of a Kalman filter?

A Kalman filter is a popular tracking method for predicting the motion of objects in an image. It uses a motion model for that system and a measurement of the state to predict the next state.

9. To use a Kalman filter, what are the requirements/assumptions about the system that is being tracked?

- a. The system update is linear
- b. Noise is gaussian
- c. Error criteria is quadratic (i.e. squared error)

10. Find a recent (after 2022) computer vision paper that makes use of a Kalman Filter along with a deep network.

<https://arxiv.org/pdf/1911.03112>

This paper details how the trajectory of an object pushed across a flat plane can be predicted. A core feature is that an EKF is used to predict the system parameters like surface friction by comparing the predicted object motion / pose from an analytical model and the actual observed object pose from a vision system. A deep network is used to compute the surface normals for used in the analytical pushing model.